

2.2 Evaluating the Clarity of an Algorithm

وضاحت (Clarity) کا مطلب ہے الگورتھم پڑھنا، سمجھنا اور بدلنا آسان ہو۔

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 05 of 24

2.2 Evaluating the Clarity of an Algorithm

LEARN IT

Clarity means an algorithm is easy to **read, understand and modify**. In many cases, evaluating clarity is even more important than making the algorithm slightly faster. A clear algorithm is easier to **maintain, debug and extend** over time.

The two key aspects of clarity are:

Clarity → Modularity (parts) → Readability (text and flow)

WORKED EXAMPLE

Worked example: speed versus clarity. Algorithm A finds the class average in 8 confusing lines with variables a , b , c . Algorithm B uses two named functions and takes the same number of additions. Both are $O(n)$. Algorithm B is the better school solution because the next person (or you, in three months) can change it without breaking it.

REMEMBER IT

Ask two questions: **"Can I split this into parts?"** (modularity) and **"Can another student read it aloud and understand it?"** (readability).

COMMON MISTAKE

Mistake: treating a clever one-line trick as "better" than a clear five-line solution.

Fix: the source says clarity is often more important than speed. Board answers should praise meaningful names and modules, not compacted code.

اردو میں سمجھیں

وضاحت (Clarity) کا مطلب ہے الگورتھم پڑھنا، سمجھنا اور بدلنا آسان ہو۔ بہت سی صورتوں میں وضاحت رفتار سے بھی اہم ہے۔ اس کے دو حصے ہیں: ماڈیولریٹی (Modularity) یعنی کام کو حصوں میں بانٹنا، اور پڑھائی پن (Readability) یعنی متن اور بہاؤ صاف ہونا۔

CHECK YOURSELF

1. What does clarity mean?
2. Name the two key aspects of clarity.
3. Why can clarity matter more than a small speed gain?

Answers: 1. The algorithm is easy to read, understand and modify. 2. Modularity and readability. 3. Because a clear algorithm is easier to maintain, debug and extend over time.

Next lecture: 2.2.1 Modularity